

FIFO in the datapath

UART bytes arrive asynchronously from the line rate while software reads when it can

Starter depth four

- Starter preset: depth-four FIFO, empty, write and read pointers both at zero, count zero
- Din is hex A5
- Click Push once
- Slot zero holds A5
- Pop once and you get A5 back
- Fill with four pushes to see full equal one and a fifth push blocked in the event log

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

FIFO pointer & flags

Step a small synchronous FIFO: write/read pointers, occupancy count, and empty / full flags. Push and pop to see the circular buffer move. Starter: depth-4 empty FIFO — push 0xA5.

Starter example: depth-4 empty FIFO. Push 0xA5 — count becomes 1, empty clears, wr advances.

Challenge 0 / 22 cleared

Quiz: FIFO: FIFO means...

☐ First-In First-Out — oldest entry is read next

☐ First-In Last-Out (a stack)

☐ random access like RAM

☐ only a clock generator

Show hint

Check

Next

Load challenge setup

Idle

Quiz: FIFO

Quiz: empty

Quiz: full

Quiz: write ptr

Quiz: read ptr

Quiz: push when full

Quiz: pop when empty

Quiz: sync FIFO

Push 0xA5

Fill to full

Push blocked

Pop recovers A5

Pop blocked

Write wrap

FIFO order

FIFO lab starter

Real RTL/TB practice

- In Track A, restate FIFO in one sentence and why UART RX often needs one
- Draw four slots with wr and rd pointers after one push of A5
- Write the empty and full conditions using count and depth
- Optional: peek at FIFO examples in the legacy combined materials and name where full would
- This lab is pointer literacy

Pitfalls to watch

- Do not pop when empty or push when full in real hardware without checking flags
- Write and read pointers wrap modulo depth
- A FIFO fixes overrun only if software or DMA keeps draining, it is not infinite buffering
- Almost-full at count depth minus one is a common interrupt threshold
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser
- On paper, sketch wr and rd after two pushes and one pop
- When you are ready, take the short quiz, then continue to handshake